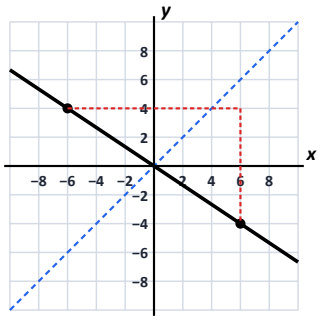
Exit Ticket Answer Key

Module 2, Topic 2, Lessons 1–3: Transformations, Parallel and Perpendicular Lines · question n is modeled on End of Topic Assessment item n

1. f(x) = x, m(x) = a · f(x). What is a? **a = −2/3**

1. m(x) = a · x, so a is the slope of m.
2. From (−6, 4) go right 12 and down 8: slope = −8/12 = **−2/3**. Or (−4 − 4) / (6 − (−6)) = −8/12.
3. Check: m(6) = −2/3 · 6 = −4 ✓. f(x) = x is compressed by 2/3 and reflected over the x-axis.

2/3: missed the reflection (line falls). **−3/2:** did run over rise.



2. Range for $40 + $15 per hour, at most 5 hours. **B. 40 ≤ y ≤ 115**

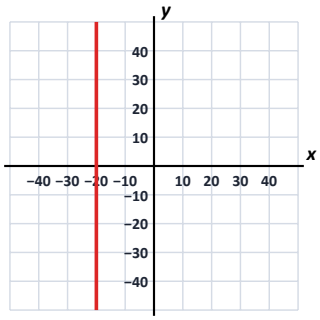
1. y = 15x + 40 with 0 ≤ x ≤ 5 (the domain).
2. Smallest cost: x = 0 → y = 40. Largest cost: x = 5 → y = 15(5) + 40 = 115. Range: **40 ≤ y ≤ 115**.

C: that is the domain (x), not the range (y). **A:** started at the hourly rate instead of the fixed fee.

3. Equation and slope of the line. **F. x = −20; slope is undefined**

1. Each grid line is 10. The line is **vertical** through x = −20, so every point has x = −20.
2. Vertical line: run = 0, and you can't divide by 0, so the slope is **undefined**. (Horizontal y = c has slope zero.)

G/J: mixed up x = and y =. **H:** mixed up undefined (vertical) and zero (horizontal).



4. g(x) = f(x − 3). **C. translated right 3 units**

1. −3 is **inside** the parentheses → inputs change → left/right. Inside works opposite the sign: − inside moves **right**.

B: treated −3 as outside (down). **D:** knew it was horizontal but went the way the sign looks.

5. Translate f(x) 2 units right. y-intercept? **(0, 7)**

1. Right 2: every x-value + 2, y-values stay. x = 0 on g came from x = −2 on f, and f(−2) = 7.
2. Check: f(x) = −2x + 3, g(x) = f(x − 2) = −2(x − 2) + 3 = −2x + 7.

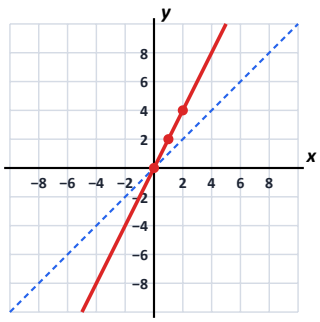
(0, −1): moved left. **(0, 5):** moved up 2. **(0, 3):** the original y-intercept.

x	f(x)	x + 2
−2	7	**0**
−1	5	1
0	3	2
1	1	3

6. f(x) = x, g(x) = f(2x). **g(x) = 2x**

1. Replace x with 2x: g(x) = 2x. Plot (0, 0), (1, 2), (2, 4).
2. Horizontal compression by 1/2: the line is steeper than f(x).

y = x/2: stretched instead of compressed. **y = x + 2:** added instead of multiplied.



7. Perpendicular to $y = -3/2\,x + 1$, crosses the x-axis at 6. **$y = 2/3\,x - 4$**

1. Negative reciprocal of $-3/2$: flip → $-2/3$, change the sign → **2/3**. Point: **(6, 0)**.
2. $y - 0 = 2/3(x - 6)$ → $y = 2/3\,x - 4$. Check: $2/3 \cdot 6 - 4 = 0$ ✓

> **−2/3:** flipped, forgot the sign. **3/2:** changed the sign, didn't flip. **(0, 6):** used 6 as the y-intercept.

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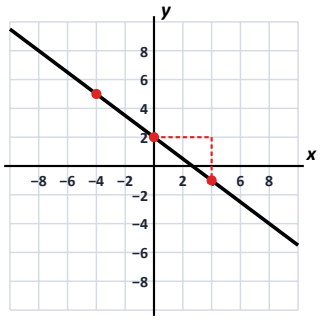
8. Parallel to $y = 1.5x + 4$, through $(-2, 0)$. **$y = 1.5x + 3$**

1. Parallel → same slope 1.5. $y - 0 = 1.5(x + 2)$ → $y = 1.5x + 3$. Check: $1.5(-2) + 3 = 0$ ✓

> **1.5x − 3:** sign error with $x - (-2)$. **1.5x + 4:** kept the old y-intercept.

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9. Point-slope form of the graph. **$y - 2 = -3/4(x - 0)$**

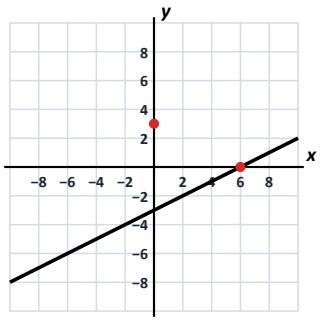


1. Find two grid points: (0, 2) and (4, −1). From (0, 2): right 4, down 3 → slope **−3/4**.
2. $y - y_1 = m(x - x_1)$ → $y - 2 = -3/4(x - 0)$.
3. Also correct: $y + 1 = -3/4(x - 4)$, $y - 5 = -3/4(x + 4)$, $y + 4 = -3/4(x - 8)$. Any point on the line works.

> **slope 3/4:** missed that the line falls. **−4/3:** run over rise. Sign errors: $y - (-1)$ is $y + 1$.

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10. Select TWO. **H and K**



1. The line rises left to right → increasing (F is false). Slope: from (0, −3) right 2, up 1 → 1/2 (G is false).
2. Crosses the y-axis at (0, −3) → **H** ✓. Crosses the x-axis at $x = 6$ → the zero is 6, **K** ✓.
3. Equation: $y = 1/2\,x - 3$. Check: $1/2 \cdot 6 - 3 = 0$ ✓

> **J:** the zero is where $y = 0$ (x-axis), not the y-intercept. **G:** rise over run flipped.

Next step: Q1, 4, 5, 6 → Lesson 1 and 2 videos. Q7, 8 → Lesson 3 video and homework. Q2, 3, 9, 10 → earlier skills: review on Mon 11/2.